A COMPLETE WALK-THROUGH · FOR EVERYONE · NO TECH KNOWLEDGE NEEDED

ArchPi

One search box. Any building on Earth. Behind it, a team of AI researchers, a real physics engine, satellite soil data, live weather, and a forensic materials lab — working together like a firm of engineers that never sleeps.

BUILT SOLO · 4 SERVERS · 11 PAGES · 119 AUTOMATED TESTS · ₹0 MONTHLY RUNNING COST

Every building is slowly dying. Almost nobody is watching.

Think about the buildings around you — your school, a hospital, a 200-year-old temple, a bridge you cross daily. Every one of them is under attack, every day: **concrete cracks, iron rusts and swells, salt eats stone from the inside, the ground underneath shifts, heat expands walls by day and cold shrinks them by night.**

Now ask: how would anyone find out if a specific building is in danger? Today the honest answer is — you hire a structural engineer, a soil-testing company, a materials laboratory, and a historian. That costs **lakhs of rupees and weeks of time**. So for most buildings, it simply never happens until something visibly breaks.

THINK OF IT LIKE THIS

A building is like a human body. It needs regular health check-ups — blood tests, X-rays, a doctor reading the reports. But imagine if a full body check-up cost ₹5 lakh and took a month. Most people would never get one. That's exactly the situation for buildings today.

The question that started ArchPi

"What if a full engineering check-up of any building could start from just typing its name — and cost nothing?"

ArchPi is the answer: it gathers the research, runs the physics, checks the soil, reads the weather, and analyses the materials — automatically, in minutes, using AI services that are free to use.

Heritage conservation · Civil engineering students · Disaster preparedness · Curious minds

What is ArchPi, in one breath

ArchPi is a website where you **type the name of any building** — the Eiffel Tower, the Taj Mahal, or the office down your street — and a team of AI specialists **researches it, draws it, stress-tests it with real physics, examines the soil beneath it, measures the weather attacking it, and analyses what it's made of.** All by itself. In minutes.

IN PLAIN WORDS

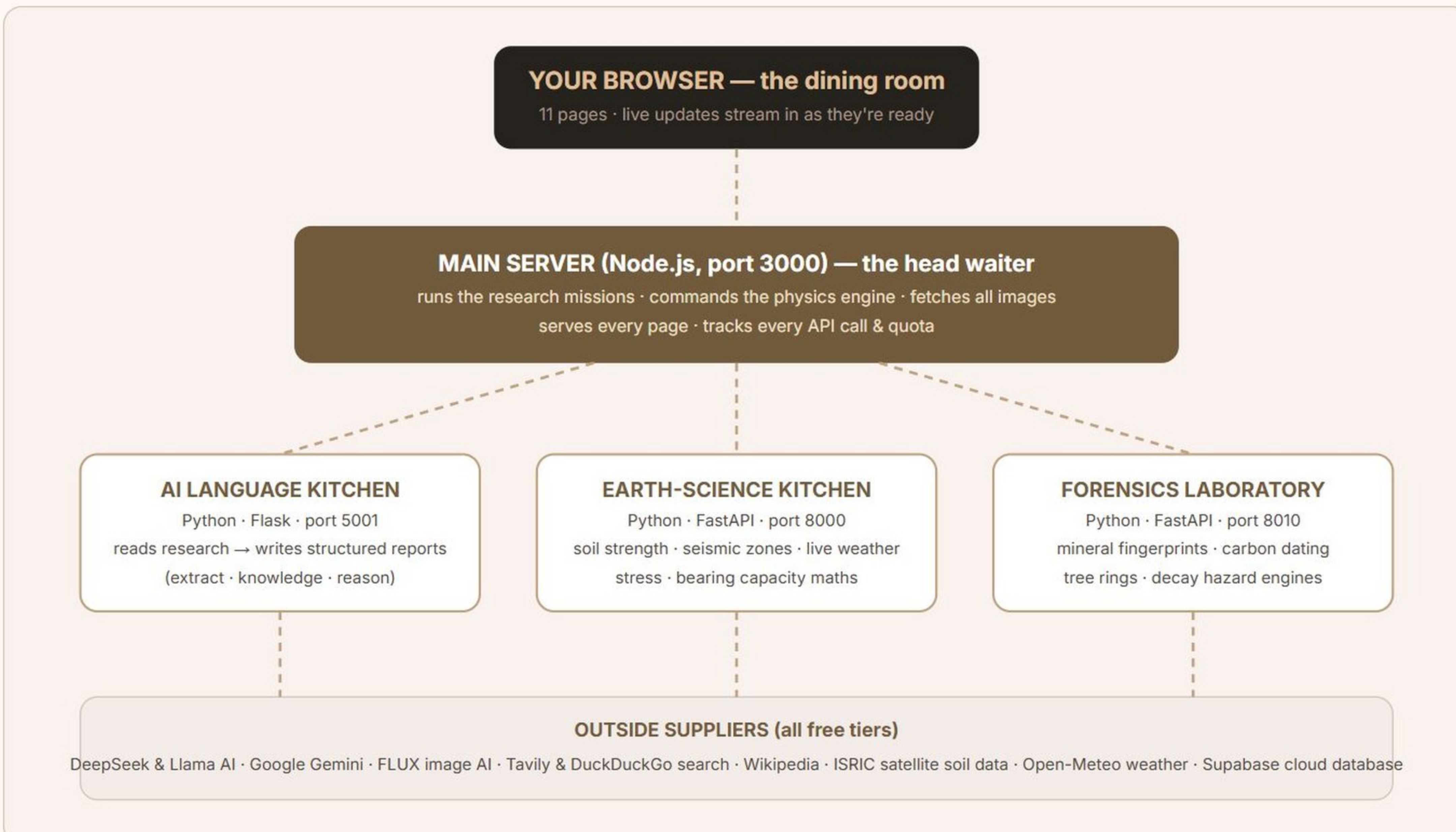
It's like having a **whole engineering consultancy firm inside your browser** — a researcher, a draftsman, a physicist, a geologist, a meteorologist and a lab scientist — except they're all software, they work in minutes, and they never send you a bill.

11 INTERACTIVE PAGES	**4** SERVERS WORKING TOGETHER	**119** AUTOMATED TESTS PASSING	**8** AI & DATA PROVIDERS
30 API ENDPOINTS (FEATURES)	**5** FALLBACK CHAINS (BACKUPS)	**500** MB FREE DATABASE SPACE	**₹0** MONTHLY RUNNING COST

How the whole machine is organised

THE RESTAURANT ANALOGY

Imagine a grand restaurant. **The dining room** is your browser — where you sit and order. **The head waiter** is our main server: he takes every order, coordinates everything, and brings the dishes out. Behind him are **three specialist kitchens** (the AI language kitchen, the earth-science kitchen, the forensics lab), and each kitchen phones **outside suppliers** — Wikipedia, satellite soil databases, weather stations — for fresh ingredients. You never see the kitchens. You just get the meal.



Why four separate servers instead of one? The same reason a restaurant has separate kitchens: if the dessert station catches fire, you can still get your main course. When one service is down or slow, the rest of ArchPi keeps working.

From a typed name to a full engineering dossier

1. **You type a building name** — say, "Colosseum" — and press Search.
The browser opens a live one-way channel (called Server-Sent Events) so you can literally watch the AI think, step by step, like watching a chef through a glass kitchen wall.

2. **Web research begins** — 5 Tavily searches + 3 DuckDuckGo searches fire at once.
They ask focused questions: overview & architect, construction timeline, materials & cost, engineering challenges, blueprint & dimensions. Roughly 40 web sources come back.

3. **The AI adds its own knowledge**.
The language model is asked: "what do you already know about this building that the web didn't say?" — facts, dates, engineering details from its training.

4. **The AI reasons over everything**.
A second AI pass cross-checks the web findings against its knowledge and resolves contradictions (two sources saying different heights, for example).

5. **The structured report is written**.
The AI turns the pile of raw text into clean, organised data: a timeline with dates, a materials list with quantities, specifications, and engineering challenges — like turning a messy pile of newspaper clippings into a neat file.

6. **Pictures are found for every section**.
DuckDuckGo image search finds a photo for each material and each timeline event; Wikipedia provides the hero shot; a blueprint image is hunted separately.

7. **Physics, on demand**.
On the Simulation page, a real Finite-Element engine (pure mathematics, no AI) stress-tests the structure under earthquake or storm loads.

8. **Everything streams to your screen**.
Findings, sources, reasoning and progress appear live as they happen — total time about 3–6 minutes for a deep report.

Why streaming matters

A 4-minute wait behind a blank spinner feels broken. The same 4 minutes spent **watching the AI find sources, state findings and explain its reasoning** feels alive — and it builds trust, because you can see where every fact came from.

ANALOGY

It's the difference between dropping your car at a garage and waiting outside a closed shutter — versus a garage with a glass wall where you watch the mechanic work. Same wait; completely different feeling.

UNDER THE HOOD (FOR THE CURIOUS)

The main server file (**server.js**) orchestrates this: **tavilySearch()** runs the web queries in parallel, **duckDuckGoTextSearch()** adds a second opinion, the LangChain agent's **/knowledge → /reason → /extract** endpoints do the AI passes, and **attachResearchImages()** pins a picture to every finding before the final report is sent.

Which AI models — and *why exactly these*

ArchPi never depends on a single AI. Every job has a **first choice, a substitute, and an emergency substitute** — chosen by balancing three things: intelligence, cost, and reliability.

THE SUBSTITUTE-TEACHER ANALOGY

A school never cancels class when a teacher is absent — a substitute walks in, then a second substitute if needed. ArchPi treats AI models the same way. The class (your request) is never cancelled.

JOB	1ST CHOICE	BACKUPS	WHY THIS CHOICE IS SMART & COST-EFFECTIVE
Reading & writing reports	**DeepSeek-V3.2** via HuggingFace router	Llama-3.3-70B → Gemini 2.5 Flash → Gemini Flash-Lite	DeepSeek-V3.2 rivals models costing 10–20× more; through HF's router it uses monthly included credits. Gemini's free tier (250 requests/day) makes the backup literally free. Paying for GPT-class APIs would cost thousands of rupees monthly for the same quality.
Understanding photos	**Gemini 2.5 Flash Vision**	— (graceful error)	Best free vision model available: identifies a building from one photo AND estimates its dimensions and shape profile. No open model matches it at ₹0.
Drawing blueprints	**Gemini image / OpenRouter**	FLUX.1-dev (HF) → Pollinations (free, no key)	Image generation is the most expensive AI job, so the chain ends at Pollinations — a completely free FLUX endpoint. Quality drops slightly; the app never stops working. This chain saved the app the day all paid quotas ran out (it happened — the app kept drawing).
Web research	**Tavily** (AI-ranked search)	DuckDuckGo (unlimited)	Tavily gives clean, AI-ready answers (1,000 free/month); DuckDuckGo is the free unlimited second opinion. Google's official search API would cost $5 per 1,000 queries.
Physics & lab maths	**No AI at all — NumPy/SciPy**	n/a	Deliberate: safety-critical numbers must be computed, never "guessed" by a language model. Pure maths is free, instant, and provably correct.

THE GOLDEN RULE OF ARCHPI

The AI is a storyteller, never a calculator. Every number that matters — stress, deflection, bearing capacity, calibration curves — comes from deterministic mathematics. The AI only reads those results afterwards and explains them in plain English. A poet is never allowed to do the surgery.

CHAPTER 6 · THE GRAND TOUR

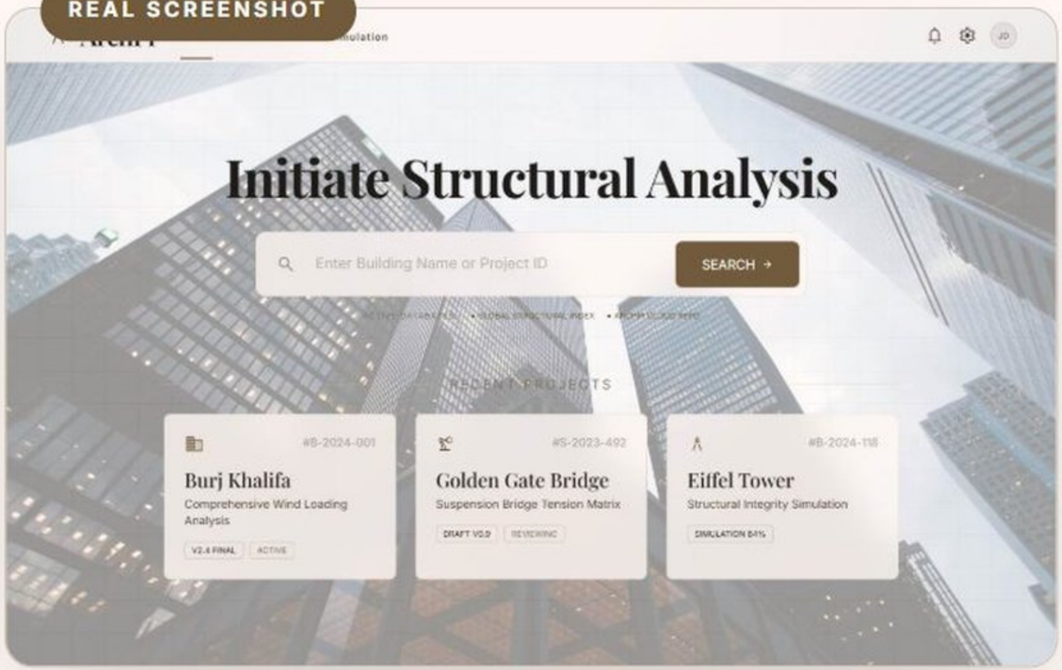
Eleven pages. Eleven real problems solved.

For each page: a real screenshot · why it exists · the real-life problem it solves · exactly how it works inside · the resources it uses · and how *you* use it. Let's walk the floor.

Home · Analysis · Simulation · Materials · Efficiency Lab · History · Soil · Weather · Forensic Lab · Globe · API Usage

Home — the front door

REAL SCREENSHOT



Why this page exists

Powerful tools fail when people don't know where to start. Home gives you exactly one decision to make: **type a building's name.** Recent analyses appear as cards so returning users jump straight back in.

The real-life problem it solves

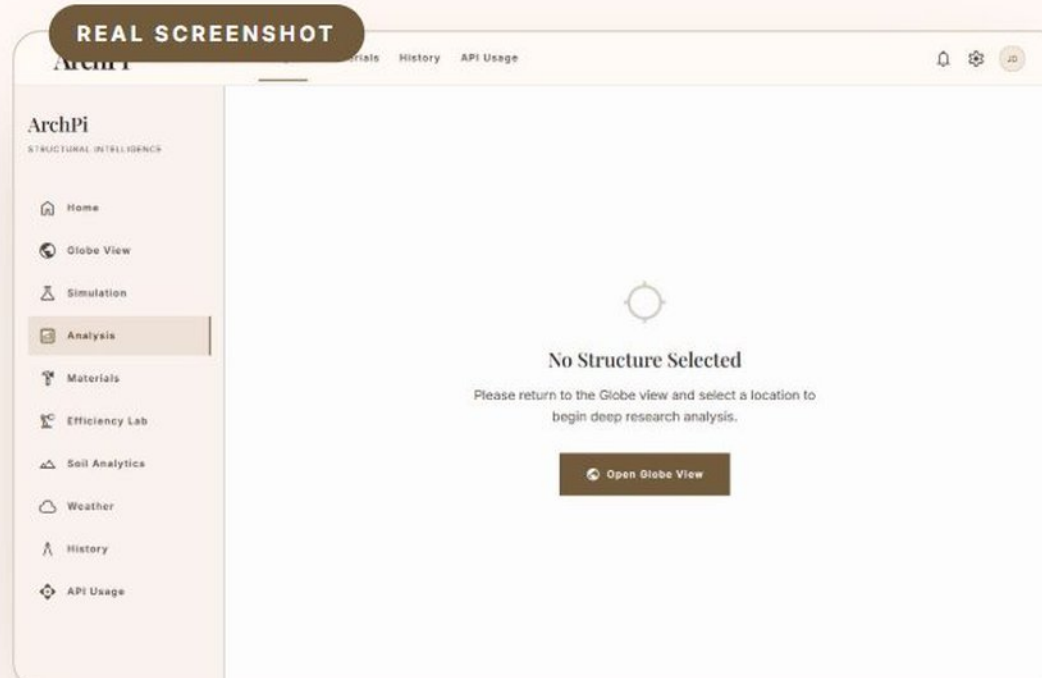
Complex engineering software normally needs training courses. ArchPi's home page needs zero explanation — if you can use Google, you can use ArchPi. That's what makes an expert tool usable by a student, a heritage officer, or a curious neighbour.

> ANALOGY
>
> It's the hospital reception desk. You don't need to know which department does an MRI — you just say the patient's name, and reception routes you to the right specialists.

Express static server · Tailwind CSS design · Cost: free

Analysis — the deep research engine

REAL SCREENSHOT



Why this page exists

Before you can judge a building, you must know it: who built it, when, with what, and what went wrong during construction. Gathering that normally means days in libraries and archives. This page does it in minutes — **and shows its working live.**

How it works inside

1. **8 web searches fire in parallel**
   5 through Tavily (an AI-focused search engine) + 3 through DuckDuckGo — about 40 sources gathered on overview, timeline, materials, challenges, blueprints.

2. **Three AI passes**
   /knowledge — the AI adds facts it already knows. **/reason** — it cross-checks and resolves contradictions. **/extract** — it writes the final structured dossier (timeline, materials, specs).

3. **Images pinned to every finding**
   Every material and timeline event gets its own photograph via image search; Wikipedia supplies the hero shot.

4. **Everything streams live**
   Server-Sent Events push each phase, source, finding and thought to your screen the moment it happens.

ANALOGY

You hire a team of six librarians. One reads newspapers, one reads engineering journals, one interviews an old professor (the AI's own memory), one fact-checks them against each other, one types the final report, and one glues a photo next to every paragraph. ArchPi's team just happens to finish in four minutes.

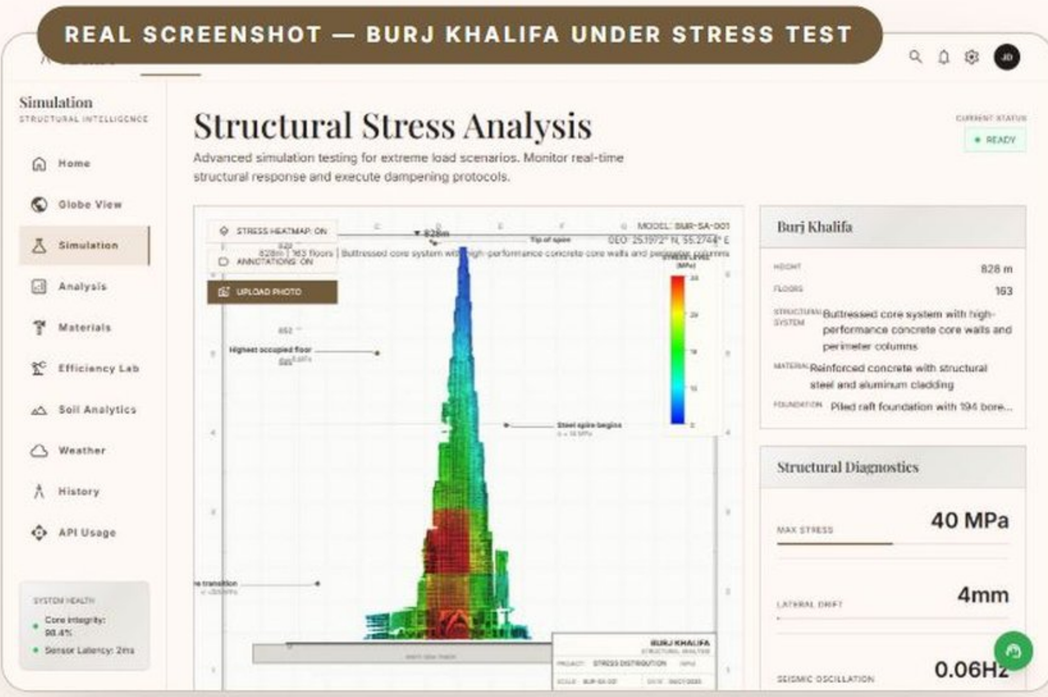
REAL-LIFE IMPACT

A civil-engineering student preparing a case study on the Brooklyn Bridge gets a sourced, illustrated dossier in one coffee break — work that used to take a **week**.

Tavily search · DuckDuckGo · DeepSeek-V3.2 · Gemini fallback · Wikipedia images · Server-Sent Events · 1,000 searches/mo free

Simulation — a virtual earthquake machine

REAL SCREENSHOT — BURJ KHALIFA UNDER STRESS TEST



Why this page exists

"Would this building survive an earthquake?" is the single most important question in structural engineering — and normally answering it needs software licences costing lakhs (ETABS, ANSYS) plus an expert to drive them.

How it works inside — real physics, zero AI guessing

1. **The AI builds the profile**
   `/api/building-profile` asks the language model for the building's real height, materials and silhouette — the only part AI does.

2. **A skeleton is generated**
   The structure becomes a frame of nodes and beams — like a matchstick model of the building.

3. **Pure mathematics takes over**
   fea_engine.py runs the Finite Element Method (the direct stiffness method): a large matrix equation K·u = F is solved with NumPy. This is textbook engineering maths — the same family of method used by professional analysis software.

4. **Results are painted & narrated**
   Stresses colour the drawing (blue = relaxed, red = screaming). DeepSeek then reads the raw numbers and writes the engineer's log in plain English — interpretation only, after the maths is done.

ANALOGY

Universities test buildings by putting scale models on a **shake table** — a platform that reproduces earthquakes. This page is a shake table made of mathematics: same idea, no concrete dust, and it fits in your browser.

PROOF IT'S REAL MATHS

The engine ships with a verification suite: it simulates a standard cantilever beam and compares against the 200-year-old textbook formula. The computed answer matches theory to **17 decimal places**. An AI "guess" could never do that. Bonus: upload a **photo** of any building — Gemini Vision identifies it, FLUX redraws it as a blueprint, and the physics runs on that.

fea_engine.py (NumPy) · Direct stiffness method · Gemini Vision (photo ID) · FLUX blueprint drawing · WebSocket live log · Physics: free forever

Materials — the building's recipe card

Why this page exists

A building is a recipe: the Eiffel Tower is 7,300 tonnes of wrought iron held by 2.5 million rivets; the Taj Mahal is white Makrana marble over a brick core. Knowing the ingredients tells you how it ages, what repairs need, and what it would cost today.

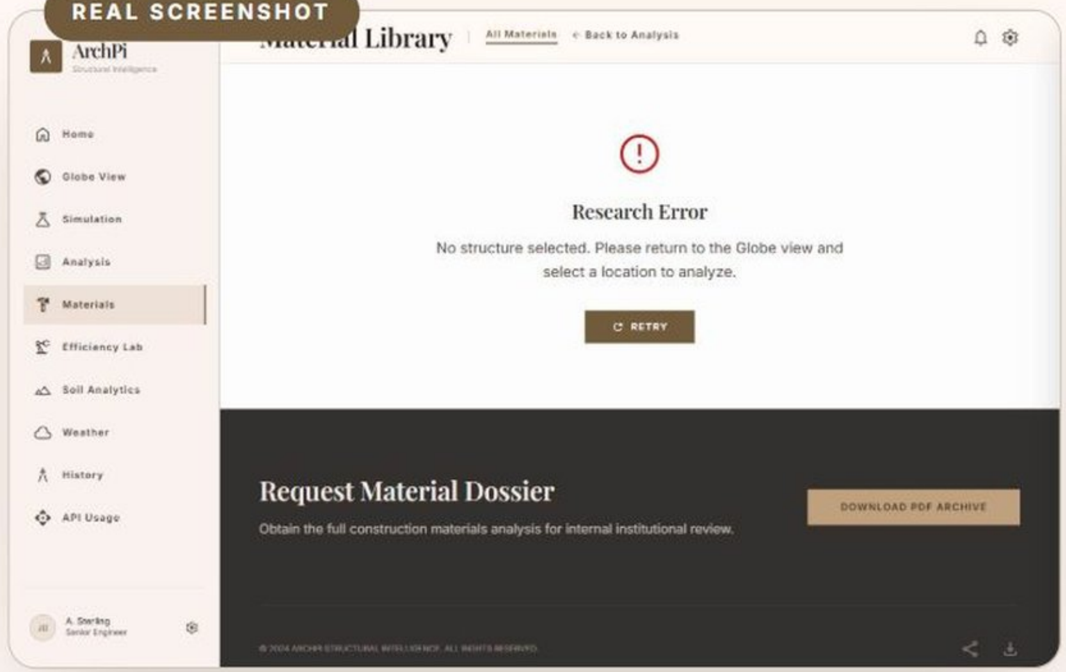
How it works & what it solves

The research engine's extraction pass produces a structured materials list — name, quantity, role, sourcing, cost — and **attachResearchImages()** finds a real close-up photo of each material *on that very building*. Restorers use exactly this kind of sheet to source matching stone; students use it to understand why iron was revolutionary in 1889.

> ANALOGY
>
> It's the ingredients label on a food packet — but for a monument, with a photo of every ingredient and the story of where it was bought.

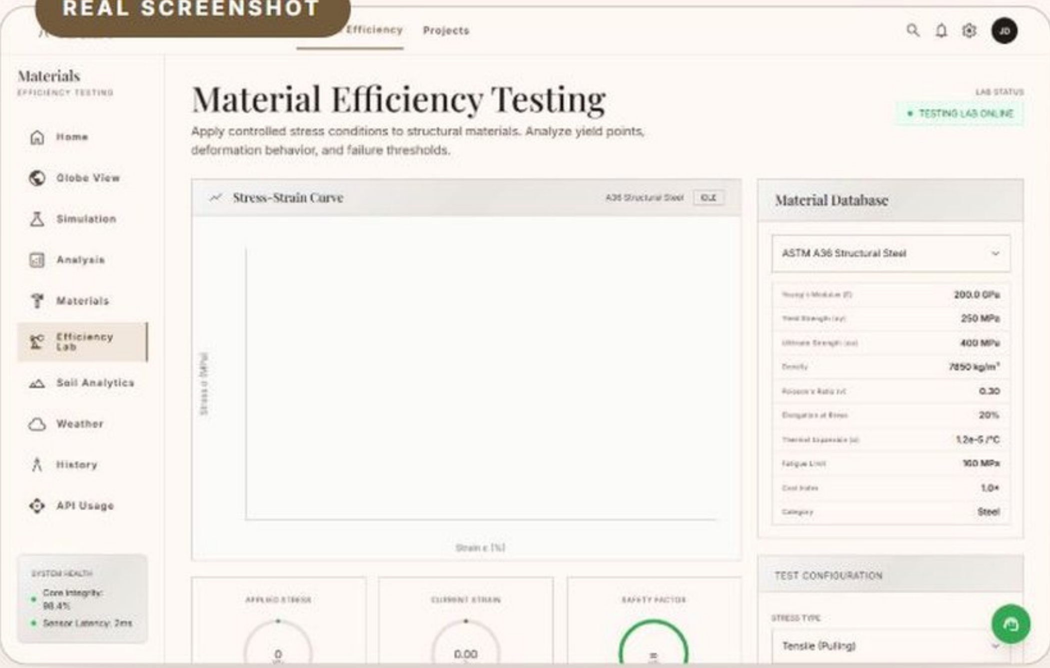
AI extraction · DuckDuckGo images · Structured JSON schema · Free tier

REAL SCREENSHOT



Efficiency Lab — could we build it better today?

REAL SCREENSHOT



Why this page exists

Construction produces roughly **a third of the world's carbon emissions**. The most powerful climate question in engineering is: "if we built this again today, what should change?"

How it works & what it solves

The lab takes the building's materials list and pits each material against modern alternatives — analysing strength-to-weight, cost and embodied carbon, with the AI explaining the trade-offs in plain terms. It turns a vague sustainability debate into side-by-side numbers.

ANALOGY

It's a dietician for buildings: "same strength, half the carbon — swap this steel for engineered timber here, keep the concrete there." A diet plan, but for a skyscraper.

Materials database · AI comparison engine · Efficiency scoring · Free tier

History — the building's life story

Why this page exists

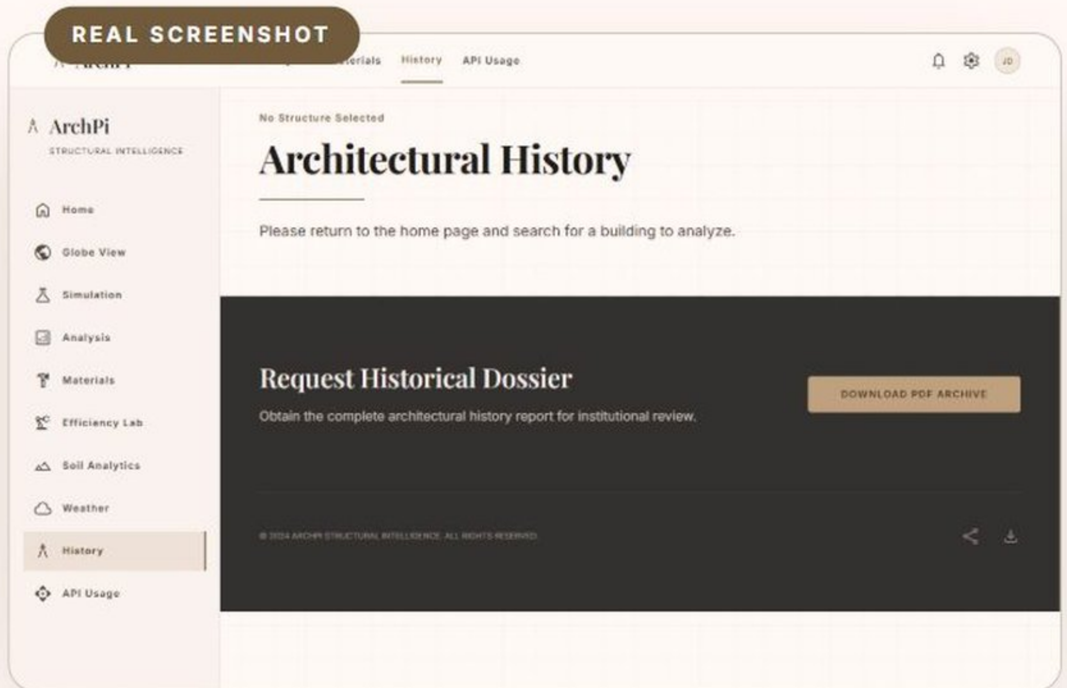
You cannot diagnose a patient without their medical history. A building that survived an 1897 earthquake, a 1944 fire and a 1980 renovation carries scars and strengths from each event — engineers call this the structure's *loading history*.

How it works & what it solves

The research engine's timeline extraction is rendered as an elegant scrolling story: every phase dated, described, and illustrated with a period-correct photograph found by the image agents (they search "*building + event + year + historical photo*"). Heritage bodies need exactly this dossier when applying for conservation funding.

> **ANALOGY**
>
> It's the building's Instagram feed — if it had been posting since 1632. Every major life event, in order, with pictures.

AI timeline extraction · Period photo search · Story-scroll UI · Free tier

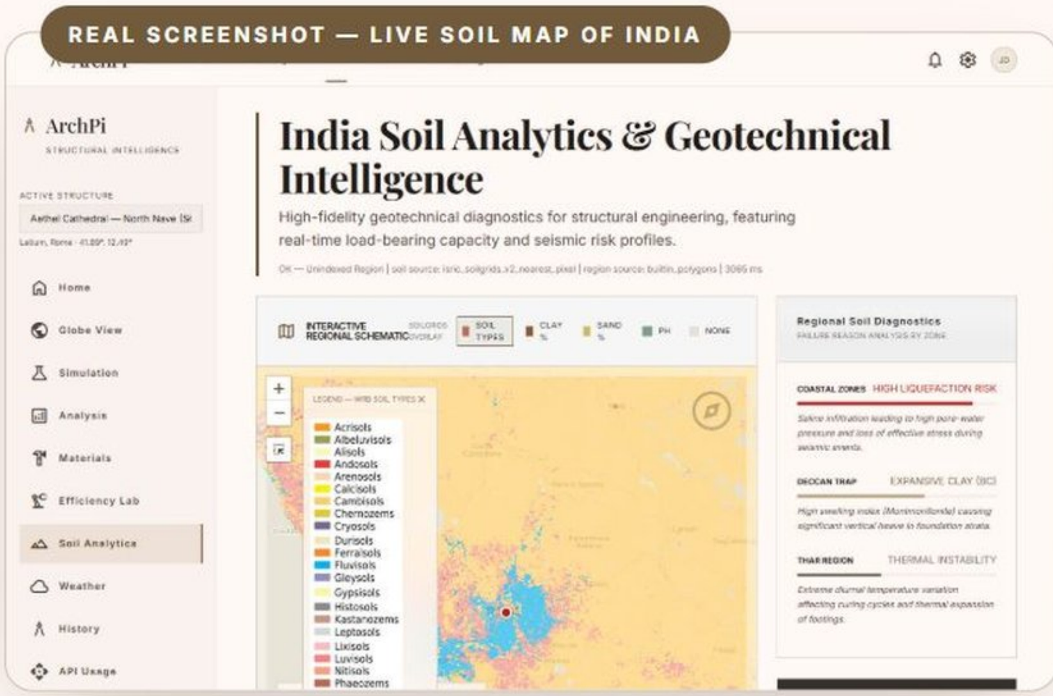


Soil Analytics — the ground doctor

Why this page exists

The most dangerous part of a building is the part you can't see: the ground. Buildings in Mexico City sank because of soft lake clay; apartment towers have toppled over *intact* because soil liquefied in an earthquake. In India — black cotton soil that swells in monsoon, desert sand, Himalayan slopes — this is a daily engineering battle.

REAL SCREENSHOT — LIVE SOIL MAP OF INDIA



How it works inside

1. **Click anywhere on the map**
   A Leaflet map of India with ISRIC soil-type rasters overlaid. Your click sends the exact coordinates to the earth-science server.

2. **Satellite soil data is fetched**
   ISRIC SoilGrids (a global scientific soil database) returns the clay %, sand %, silt %, density and pH at six depths for that exact spot.

3. **A 100-year-old formula does the maths**
   Terzaghi's bearing-capacity equation — the foundation formula of civil engineering since 1943 — computes how many tonnes per square metre that ground can safely carry. PostGIS (a map-aware database on Supabase) identifies the seismic zone.

4. **You get a verdict**
   Safe bearing capacity, foundation recommendation, liquefaction risk, seismic zone — the numbers a soil-testing company charges ₹50,000+ to produce a first estimate of.

ANALOGY

Would you stand a heavy cupboard on a soft mattress? Same cupboard, on a concrete floor — no problem. This page tells you, for any spot in India, whether the ground is mattress or concrete — before you build the cupboard.

CLEVER ENGINEERING DETAIL

City-centre clicks sometimes land on "urban" pixels where the satellite has no soil reading. The backend detects this and automatically **samples a ring of nearby points** and averages them — so you always get an answer instead of an error.

ISRIC SoilGrids satellite data · Terzaghi equation · Supabase PostGIS · Leaflet maps · FastAPI · All free tiers

Weather — the slowest demolition crew on Earth

Why this page exists

Weather never takes a day off. Wind pushes on walls, humidity seeps into concrete and rusts the steel inside (rust expands 6× and cracks the concrete open — engineers call it *carbonation-driven corrosion*), and every hot day / cold night cycle flexes the whole facade.

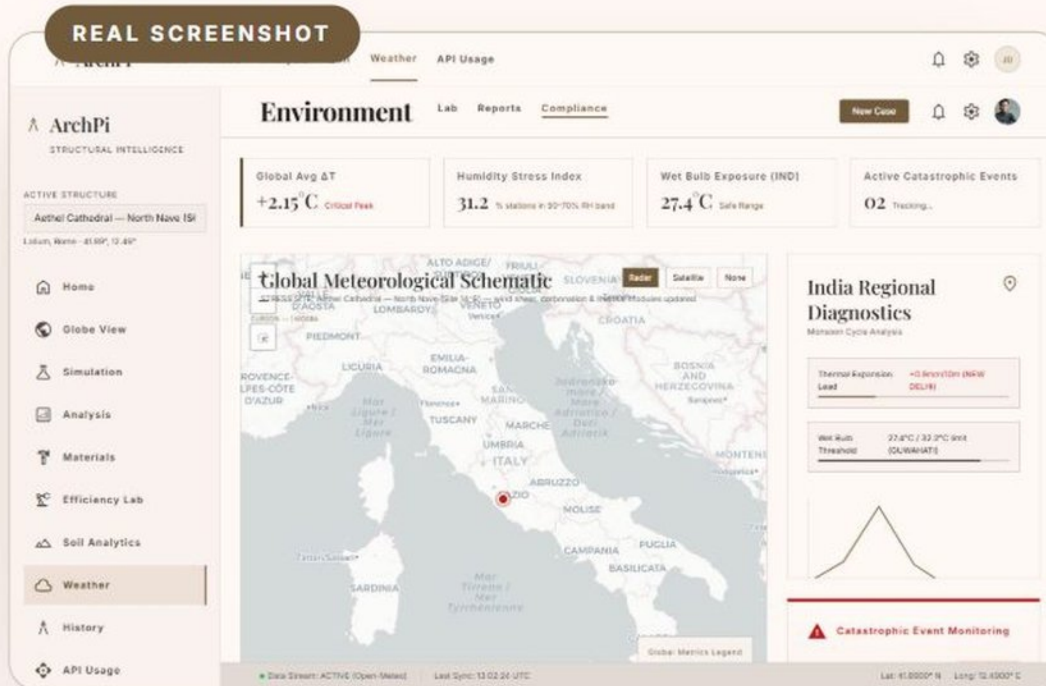
How it works & what it solves

The earth-science server pulls **live conditions from Open-Meteo** (a free scientific weather service) and 30-year ERA5 climate normals, then converts them into *engineering* numbers: dynamic wind pressure on a wall (kN/m²), a carbonation risk multiplier, thermal swing in degrees. Every reading is also logged into the Supabase time-series ledger — building a history for trend analysis. A maintenance manager sees not "it's humid today" but "**this humidity is ageing your concrete 1.8× faster than normal**".

ANALOGY

A fitness tracker for a building. Your watch turns heartbeats into health advice; this page turns weather into structural-health advice.

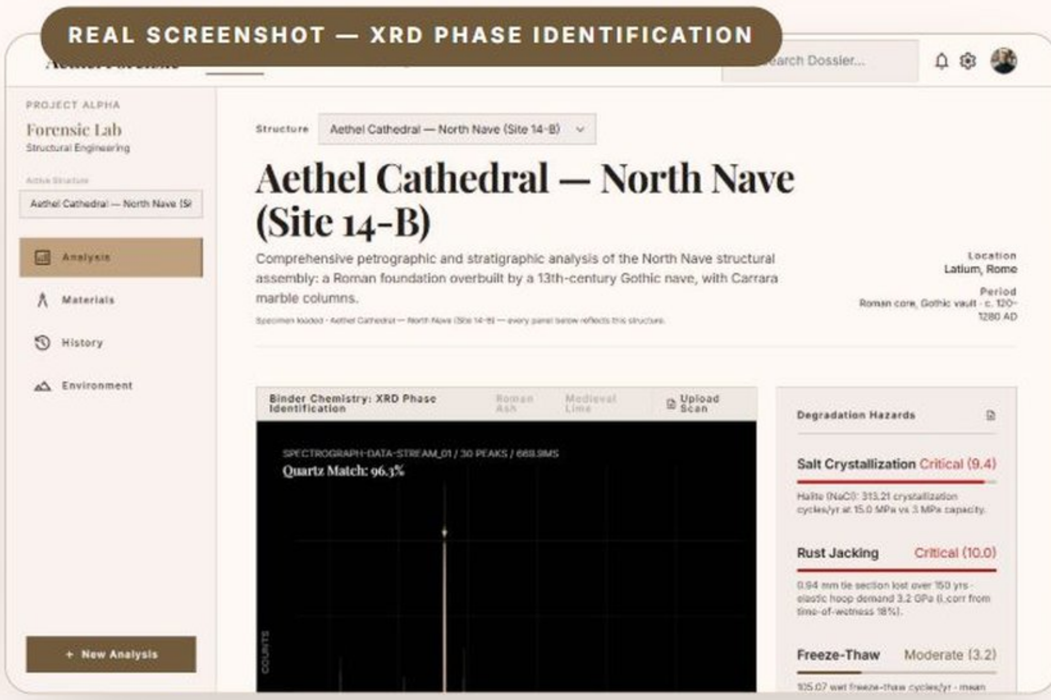
REAL SCREENSHOT



Open-Meteo live API · ERA5 30-year normals · Wind-shear formula · Supabase ledger · Free · 10,000 calls/day

Forensic Lab — CSI for buildings

REAL SCREENSHOT — XRD PHASE IDENTIFICATION



Why this page exists

Old buildings keep secrets: *When was this wing really built? Is this "original" column actually a later repair? Which quarry did this marble come from? What is eating this facade?* Real forensic labs answer these with machines costing crores. This page implements the **mathematics of those machines** in software.

The six instruments

INSTRUMENT	PLAIN-LANGUAGE JOB
XRD analysis	**Mineral fingerprinting.** Every mineral bends X-rays at signature angles — like a barcode. The engine matches peaks to identify what a mortar is made of.
Radiocarbon ^{14}C	**The carbon clock.** Living things absorb carbon-14; after death it decays at a known rate. Measure what's left → get the age, with proper statistical calibration.
Dendrochronology	**Tree-ring barcode.** Rain patterns make unique ring sequences. Match a beam's rings to the master record → the exact year the tree was felled.
Isotope provenance	**The marble passport.** Each quarry's stone has a unique chemical signature — the engine traces a sample back to Carrara vs Paros vs Makrana.
Hazard engines	**Decay forecasting.** Salt crystallisation cycles, rust-jacking of iron ties, freeze-thaw — computed from real humidity data into "critical / moderate" scores.
Stratigraphy	**Layer detective.** "This wall cuts that floor, so it's younger" — formalised as a network graph (a Harris matrix) that sorts the whole site's history.

ANALOGY

It's a crime-scene lab where the victim is a 900-year-old cathedral. Fingerprints (XRD), time of death (radiocarbon), the suspect's travel history (isotopes) — every classic CSI move, done on stone and mortar.

WHY YOU CAN TRUST IT

This is the most heavily engineered part of ArchPi: **119 automated tests** check the science against known answers on every change. The demo dossier follows one fictional cathedral ("Aethel Cathedral") through all six instruments so the story connects.

NumPy + SciPy science · NetworkX graphs · FastAPI · 119 pytest tests · No external APIs needed

Globe — the world at a glance

Why this page exists

Governments and heritage bodies don't manage one building — they manage *hundreds*. The Globe zooms out: monitored structures and environmental stress vectors across the map, so you think in portfolios, not single sites.

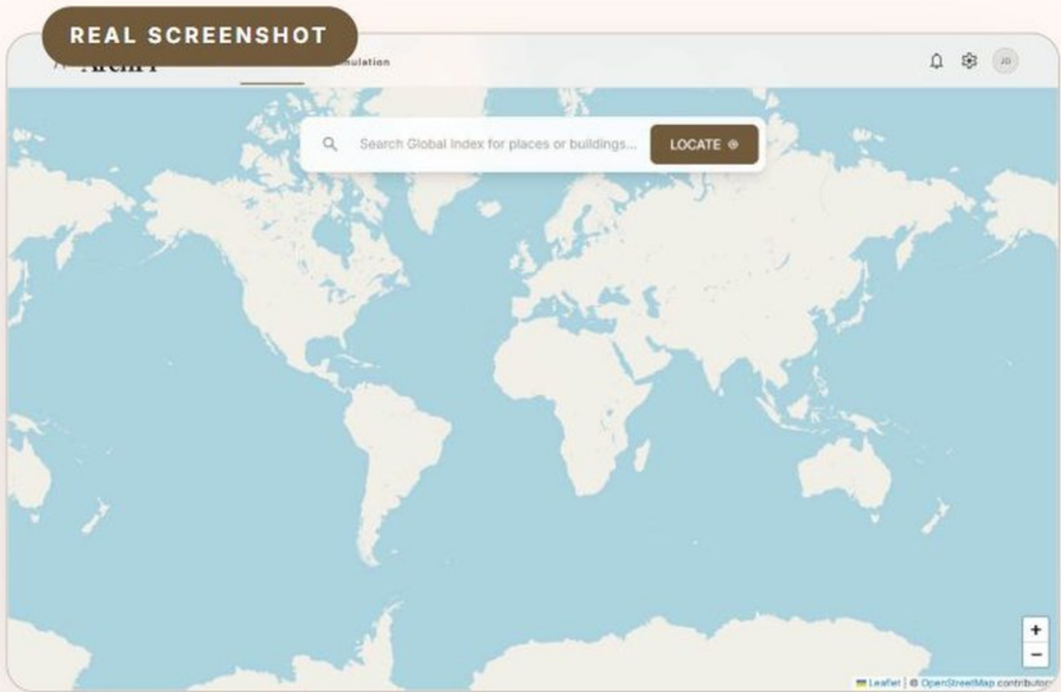
What it solves

Priority. With limited budget, which of your 300 heritage sites needs attention *first*? A world view with stress data answers that in one glance — the difference between firefighting and planning.

ANALOGY

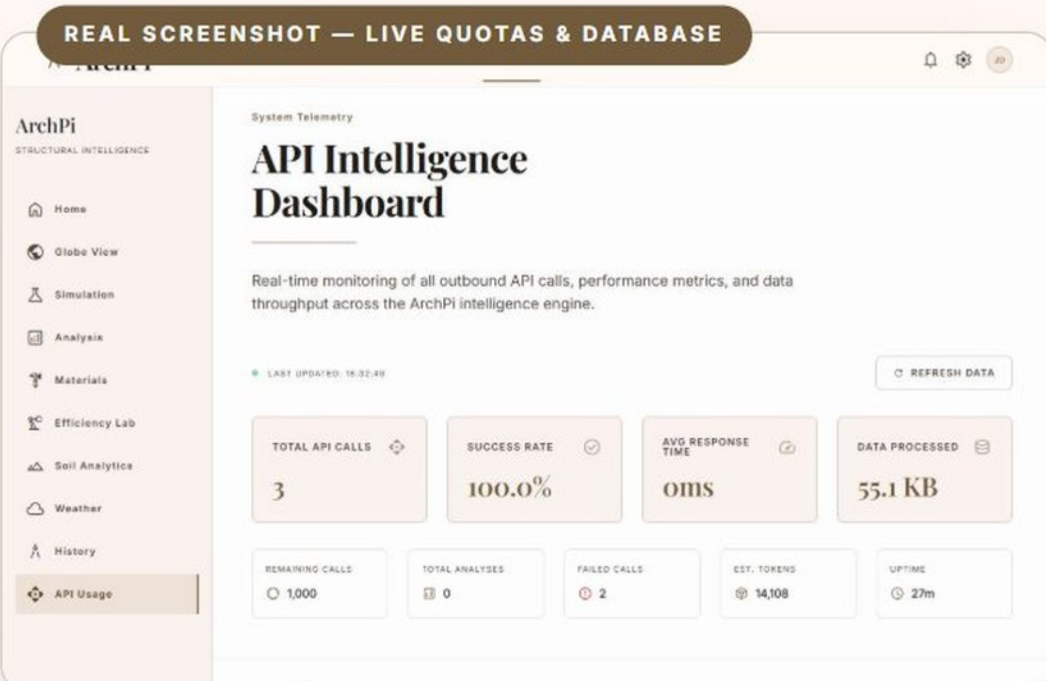
Air-traffic control for buildings. A controller doesn't study one plane at a time — they watch the whole sky and act where the risk is.

Interactive globe | Global stress vectors | Open-Meteo grid | Free tier



API Usage — the cockpit fuel gauges

REAL SCREENSHOT — LIVE QUOTAS & DATABASE



Why this page exists

Running on free services is ArchPi's superpower — and its risk, because **free means limits**: 1,000 searches a month here, 250 AI requests a day there, 500 MB of database space. Run out mid-demo without warning and the app "mysteriously" breaks.

What it shows (live)

1. **A fuel gauge per provider**
   Tavily, Gemini, HuggingFace, OpenRouter, DuckDuckGo, Pollinations — each with used vs remaining today/this month, and a red "quota exhausted" badge with the exact error if one runs dry.
2. **The live money balance**
   For OpenRouter, the dashboard calls the provider's own billing API and shows the real balance in dollars — not an estimate.
3. **Database storage meter**
   Live from Supabase: MB used of the 500 MB free plan, plus every table with its row count and size — you can see exactly what's stored.
4. **Permanent history**
   Every call is logged to **usage_data.json** on disk, so counters survive restarts — a real calendar day means a real calendar day.

> ANALOGY
>
> A car dashboard. You don't wait for the engine to stop to learn you're out of petrol — the gauge warns you 100 km before. This page is that gauge for every resource the app depends on.

Persistent counters · Live OpenRouter billing API · Supabase storage query · Per-provider quota map · Free to run

Everything ArchPi is made of — the full list

INGREDIENT	WHAT IT IS, IN PLAIN WORDS	WHAT ARCHPI USES IT FOR	COST
Node.js + Express	The language and toolkit of the main server	The "head waiter": serves pages, orchestrates every mission	**Free**
Python 3	The language scientists love	All four science/AI helpers (agents, FEA, both backends)	**Free**
FastAPI ×2 + Flask	Toolkits for building web services in Python	The earth-science kitchen, forensics lab, and AI language kitchen	**Free**
NumPy + SciPy	The world's standard mathematics libraries	The FEA physics engine and all six forensic instruments	**Free**
NetworkX	A library for relationship maps (graphs)	Sorting archaeological layers into a timeline (Harris matrix)	**Free**
LangChain	A toolkit for talking to AI models in structured ways	The extract / knowledge / reason pipeline	**Free**
DeepSeek-V3.2 & Llama-3.3	Open AI language models (via HuggingFace router)	First-choice brain: reading research, writing reports, narrating physics	**~Free credits**
Google Gemini 2.5 Flash	Google's fast AI model, with vision	Backup brain + identifying buildings from photos	**Free · 250/day**
FLUX.1-dev + Pollinations	AI image generators	Drawing CAD-style blueprint elevations of any building	**Free fallback**
Tavily + DuckDuckGo	Web search engines with APIs	The 8-query research barrage; image hunting	**Free · 1,000/mo + unlimited**
Wikipedia API	The world's encyclopedia, machine-readable	Hero photos and reference images	**Free**
ISRIC SoilGrids	A global scientific soil database (satellite + surveys)	Real clay/sand/pH data for any coordinate on Earth	**Free**
Open-Meteo	A free scientific weather service	Live weather + 30-year climate normals → building stress	**Free · 10,000/day**
Supabase (PostGIS)	A cloud database that understands maps	Seismic region lookups; weather history ledger; 500 MB free	**Free · 500 MB**
Leaflet + Tailwind CSS	Map display and design toolkits	The interactive India soil map; the entire visual design	**Free**
pytest + Puppeteer	Automated testing tools	119 science tests + robot-browser checks of every page	**Free**

Total monthly bill for all of the above: **₹0** — by design, through fallback chains and careful quota engineering. That is the "cost-effective" story: not cheap components, but *smart arrangement* of world-class free ones.

Five rules that make it engineering, not just a demo

1 · AI never does the math

Stress, deflection, bearing capacity, dating curves — all computed by deterministic formulas and verified against textbook answers. The AI only explains results afterwards. **A storyteller is never handed the scalpel.**

2 · Everything has a backup

Five fallback chains: language AI (4 models deep), image AI (3 providers), search (2 engines), region lookup (cloud → built-in), soil pixels (point → ring sampling). The day every paid quota ran out, **the app kept working**.

3 · Everything is measured

Every outbound call logged and persisted; quotas tracked against real calendar windows; live billing balances; database space monitored. **You cannot fix what you cannot see.**

4 · Everything is tested

119 automated tests pin the science to known answers; a robot browser (Puppeteer) opens every page and reports errors; the FEA ships with a verification suite matching theory to 17 decimal places.

5 · Security basics respected

Python helpers are launched without a shell (no command injection through building names); API keys live in an environment file that never enters version control; images are proxied server-side.

The philosophy

Features impress for a minute. **Failure-handling impresses forever.** Every design choice above assumes something WILL go wrong — and makes sure the user never notices when it does.

THE RECAP

What you just walked through

4	COOPERATING SERVERS
11	PAGES · 11 PROBLEMS SOLVED
119	TESTS GUARDING THE SCIENCE
6	FORENSIC INSTRUMENTS
8	AI & DATA PROVIDERS
5	FALLBACK CHAINS
17	DECIMAL PLACES OF FEA ACCURACY
₹0	MONTHLY RUNNING COST

"Type a building's name.
Get an engineering firm."

ArchPi — researched by AI, computed by mathematics, checked by tests, powered by free tiers, and explained — hopefully — so that anyone can understand it.

BUILT BY SAHITYA KATARIYA · PRESENTATION CRAFTED WITH CARE · 2026